👩🏾‍🏫 Staff Guide · Horticulture 1 · Unit 5 Soil & Nutrients (Level 3)

Level 3 = resource-room students who can work close to the gen ed lesson. No slide decks: students follow the class teacher's lesson and slides with Level 3 readings, guided notes and a quick check.

1. How the pieces line up

Gen ed lesson	Level 3 piece
Her readings and worksheets	Level 3 reading (5 short paragraphs) + check questions
Class notes	Level 3 guided notes + key
Unit quiz / test	Level 3 quick check + key
Review (Kahoot / Jeopardy)	1 Level 3 game on the class site: 10 questions + a matching round

2. Supports built in

- 🔊 Read aloud on the reading and notes pages (tap the button)
- Key terms listed at the top of the notes
- Short-answer lines sized for 1-2 sentences

3. What to grade

- Notes and reading checks: same points as her worksheets for that lesson
- Quick check: score, then scale to her test points
- Modified grading only as the IEP allows. Check with the case manager.

4. Who to ask

- Mr. Driver (SLE): changes and IEP questions.
- Facts we could not confirm say “check with the class teacher”.

🚧 Under construction (check with the class teacher)

- **Unit 4 SE question text**: The 16 self-evaluation questions are in the textbook scan only; our quiz is team-written.
- **Plant Media and Nutrition**: Title only in the export.

⚠️ Safety: tools and plants

- **Tools**: students may use all tools (pruners, knives, floral wire, low-temp glue guns) **with an adult right there**. Staff hand tools out, stay at the station, and count them back in. Cut-resistant gloves if possible.
- **Poinsettia sap** is milky and can irritate skin and eyes. Gloves on, no rubbing eyes, wash hands after.
- **Latex**: poinsettias are in the latex family. Students with a latex allergy wear non-latex (nitrile) gloves or skip handling; check health plans.
- **Bees and pollen**: check IEP/health plans for bee-sting or pollen allergies and know where the EpiPen is before greenhouse work.
- **Mouths**: no plants, soil, seeds, berries, fertilizer or rooting hormone in mouths. Students who mouth objects or have pica need one-to-one eyes in the greenhouse. Many plants (holly and mistletoe berries, some houseplants) are poisonous.
- **Chemicals**: rooting hormone and fertilizer with gloves, then wash hands. Pesticides and growth regulators are staff/teacher only; students never spray or handle them.
- **Greenhouse**: wet floors and hoses (slips), heat (water breaks), wet dry potting mix first (dust).

Soil & Nutrients: Guided Notes

Horticulture 1 · Unit 5 · Level 3 (closer to the class work) · goes with Plant Nutrition Definitions, Soil Analysis Parts 1-2, deficiencies

Name: _____

Use the class slides and your reading. Key words:

potting mix | *perlite* | *sand* | *silt* | *clay* | *loam* | *soil texture triangle* | *nitrogen* | *phosphorus* | *potassium* | *deficiency*

A. Media and soil

1. Fill in the table.

Particle	Size	Water
Sand		
Silt		
Clay		

B. Jar test math

2. **Percent of a layer**

> Percent = layer depth ÷ total depth × 100
> Example: sand 1.5 cm of 3 cm → 1.5 ÷ 3 × 100 = 50%

a) Silt 0.9 cm of 3 cm _____

b) Clay 0.6 cm of 3 cm _____

C. N-P-K

3. Fill in the table.

Nutrient	Job	Deficiency sign
N		
P		
K		

Soil & Nutrients: Reading

Horticulture 1 · Unit 5 · Level 3 (closer to the class work) · goes with Plant Nutrition Definitions, Soil Analysis Parts 1-2, deficiencies

Name: _____

Soil, Media and Plant Nutrients

Plants need light, water, air, the right temperature, and a growing medium that supplies nutrients. In a greenhouse, plants grow in potting mix, a blend of peat or bark with perlite or vermiculite. Potting mix drains well, holds air for roots and is free of weed seeds. Dry mix should be wetted first, because its dust can carry germs.

Garden soil is made of mineral particles of different sizes. Sand particles are the largest and let water drain quickly. Silt particles are medium-sized and feel smooth. Clay particles are the smallest; they are sticky and hold water. Loam is a balanced mix of the three and is ideal for most plants. A soil texture triangle uses the percentages of sand, silt and clay to name a soil type.

In a soil jar test, soil and water are shaken in a clear jar and left to settle. Sand settles first at the bottom, then silt, with clay on top. Measuring each layer and dividing by the total depth, then multiplying by 100, gives the percent of each particle.

The three main nutrients are listed on fertilizer bags as N-P-K. Nitrogen (N) promotes green, leafy growth; a deficiency causes yellowing of older leaves, and an excess causes soft, leafy growth with few flowers. Phosphorus (P) supports roots, flowers and seeds; a deficiency can cause purple leaves. Potassium (K) builds strong stems and disease resistance; a deficiency causes brown, scorched leaf edges.

Check your reading

1. Why do greenhouses use potting mix instead of garden soil?

2. In the jar test, why does sand settle first?

3. Name one deficiency symptom for N, P and K.

🟢 N-P-K Plant Check: Lab Sheet

Horticulture 1 · Unit 5 · Level 3 · goes with Deficiencies / Plant Nutrition GH Participation

Name: _____

Question: Can we diagnose nutrient deficiencies from leaf symptoms?

Materials

4 plants, symptom cards, fertilizer label

⚠️ Safety

- Gloves; staff measure fertilizer
- Never ingest; wash hands

Prediction

Which deficiency do you expect is most common in our greenhouse?

_____

_____

Procedure

1. Inspect old and new leaves on 4 plants.
2. Match symptoms to N, P or K.
3. Read a fertilizer label.
4. Recommend a fertilizer.

Data

Plant	Symptom	Likely deficiency
1		
2		
3		
4		

Analysis (Claim, Evidence, Reasoning)

1. Claim: what fertilizer would you recommend for plant 1? Evidence and reasoning.

🫙 Soil Jar Test: Lab Sheet

Horticulture 1 · Unit 5 · Level 3 · goes with Soil Analysis Parts 1-2

Name: _____

Question: What percent of sand, silt and clay is in our soil sample, and what is its texture?

Materials

clear jar with lid, soil, water, ruler, calculator, texture triangle

⚠️ Safety

- Gloves; wash hands
- Keep dry soil low to avoid dust
- Nothing in mouth

Prediction

Which layer do you predict will be largest?

_____

_____

Procedure

1. Add soil and record depth.
2. Add water, close, shake 10 s.
3. Settle overnight.
4. Measure each layer.
5. Calculate percents.
6. Name soil on the triangle.

Data

Layer	Depth (cm)	Percent
Sand		
Silt		
Clay		
Total		100%

Analysis (Claim, Evidence, Reasoning)

1. Claim: what is your soil's texture? Evidence: your percents. Reasoning: how this soil will drain and hold water.

Soil & Nutrients: Quick Check

Horticulture 1 · Unit 5 · Level 3 (closer to the class work) · goes with Plant Nutrition Definitions, Soil Analysis Parts 1-2, deficiencies

Name: _____

Multiple choice and short answer. You may have it read aloud.

1. Water drains fastest through…

A. Clay | B. Loam | C. Sand

2. A 10-5-5 fertilizer is highest in…

A. Nitrogen | B. Phosphorus | C. Potassium

3. Purple leaves and poor roots suggest a lack of…

A. Nitrogen | B. Potassium | C. Phosphorus

4. Loam is…

A. A balanced mix of sand, silt and clay | B. All clay | C. A fertilizer

5. Clay is 1 cm of a 4 cm settled sample. What percent is clay?

6. What happens to a plant given too much nitrogen?

🪱 Soil & Nutrients: Guided Notes · ANSWER KEY

Horticulture 1 · Unit 5 · Level 3 (closer to the class work) · goes with Plant Nutrition Definitions, Soil Analysis Parts 1-2, deficiencies

Name: _____

Use the class slides and your reading. Key words:

potting mix · *perlite* · *sand* · *silt* · *clay* · *loam* · *soil texture triangle* · *nitrogen* · *phosphorus* · *potassium* · *deficiency*

A. Media and soil

1. Fill in the table.

Particle	Size	Water
Sand	**largest**	**drains fast**
Silt	**medium**	**moderate**
Clay	**smallest**	**holds water**

B. Jar test math

2. **Percent of a layer**

> Percent = layer depth ÷ total depth × 100
> Example: sand 1.5 cm of 3 cm → 1.5 ÷ 3 × 100 = 50%

a) Silt 0.9 cm of 3 cm **30%**

b) Clay 0.6 cm of 3 cm **20%**

C. N-P-K

3. Fill in the table.

Nutrient	Job	Deficiency sign
N	**green leafy growth**	**yellow older leaves**
P	**roots, flowers, seeds**	**purple leaves**
K	**strong stems, disease resistance**	**brown scorched edges**

🔑 *Key written by our team from the class teacher's worksheets and slides, please check.*

🟢 N-P-K Plant Check: Lab Sheet · ANSWER KEY

Horticulture 1 · Unit 5 · Level 3 · goes with Deficiencies / Plant Nutrition GH Participation

Name: _____

Question: Can we diagnose nutrient deficiencies from leaf symptoms?

Materials

4 plants, symptom cards, fertilizer label

⚠️ Safety

- Gloves; staff measure fertilizer
- Never ingest; wash hands

Prediction

Which deficiency do you expect is most common in our greenhouse?
Often nitrogen.

Procedure

1. Inspect old and new leaves on 4 plants.
2. Match symptoms to N, P or K.
3. Read a fertilizer label.
4. Recommend a fertilizer.

Data

Plant	Symptom	Likely deficiency
1	**yellow older leaves**	**N**
2	**purple leaves**	**P**
3	**brown edges**	**K**
4	**healthy**	**none**

Expected results (your class data may differ).

Analysis (Claim, Evidence, Reasoning)

1. Claim: what fertilizer would you recommend for plant 1? Evidence and reasoning.
High-nitrogen fertilizer because older leaves are yellow.

🔑 *Key written by our team from the class teacher's lab sheets, please check.*

🫙 Soil Jar Test: Lab Sheet · ANSWER KEY

Horticulture 1 · Unit 5 · Level 3 · goes with Soil Analysis Parts 1-2

Name: _____

Question: What percent of sand, silt and clay is in our soil sample, and what is its texture?

Materials

clear jar with lid, soil, water, ruler, calculator, texture triangle

⚠️ Safety

- Gloves; wash hands
- Keep dry soil low to avoid dust
- Nothing in mouth

Prediction

Which layer do you predict will be largest?

Answers vary; garden soils are often silt or sand heavy.

Procedure

1. Add soil and record depth.
2. Add water, close, shake 10 s.
3. Settle overnight.
4. Measure each layer.
5. Calculate percents.
6. Name soil on the triangle.

Data

Layer	Depth (cm)	Percent
Sand	**1.5**	**50%**
Silt	**0.9**	**30%**
Clay	**0.6**	**20%**
Total	**3.0**	100%

Expected results (your class data may differ).

Analysis (Claim, Evidence, Reasoning)

1. Claim: what is your soil's texture? Evidence: your percents. Reasoning: how this soil will drain and hold water.
Example: 50/30/20 is a loam; drains moderately and holds water well.

🔑 *Key written by our team from the class teacher's lab sheets, please check.*

🪱 Soil & Nutrients: Quick Check · ANSWER KEY

Horticulture 1 · Unit 5 · Level 3 (closer to the class work) · goes with Plant Nutrition Definitions, Soil Analysis Parts 1-2, deficiencies

Name: _____

Multiple choice and short answer. You may have it read aloud.

1. Water drains fastest through…

A. Clay | B. Loam | **C. Sand**

2. A 10-5-5 fertilizer is highest in…

A. Nitrogen | B. Phosphorus | C. Potassium

3. Purple leaves and poor roots suggest a lack of…

A. Nitrogen | B. Potassium | **C. Phosphorus**

4. Loam is…

A. A balanced mix of sand, silt and clay | B. All clay | C. A fertilizer

5. Clay is 1 cm of a 4 cm settled sample. What percent is clay?

25%

6. What happens to a plant given too much nitrogen?

Soft, leafy growth with few flowers; roots can burn.

🔑 *Written by our team from her Plant Nutrition Definitions and Soil Analysis assignments, please check.*